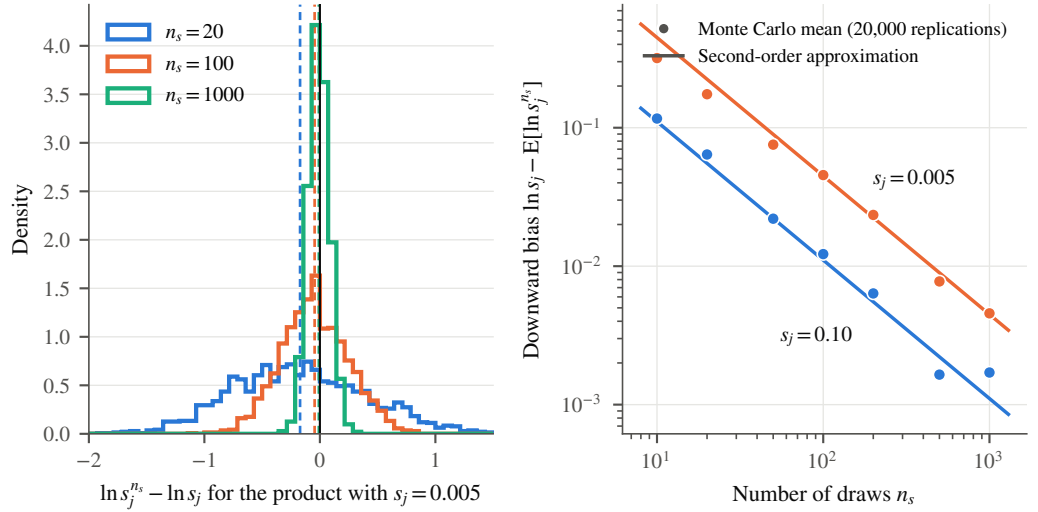
(a) Sampling distribution of the log share
$n_s = 20$
$n_s = 100$
$n_s = 1000$
Density
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−2
−1
0
1
$\ln s_j^{n_s} - \ln s_j$ for the product with $s_j = 0.005$
(b) Bias of the log share
Monte Carlo mean (20,000 replications)
Second-order approximation
Downward bias $\ln s_j - \mathrm{E}[\ln s_j^{n_s}]$
10^{-1}
10^{-2}
10^{-3}
$s_j = 0.005$
$s_j = 0.10$
10^1
10^2
10^3
Number of draws n_s